Company Visit – Themis Intelligence

Time	Duration	Session	Speaker / Organization
9:00 – 9:15	15 min	Arrival, Welcome & Introductions	**Steve Mueller**, President & CEO, Survalent; **Young Ngo**, President, Themis
9:15 – 9:35	20 min	Introduction: AIM (AI in Manufacturing); KIAT (Korea Institute of Advancement of Technology) -	**Chi-Guhn Lee**, UoT AIM Professor (AI in Manufacturing)
9:35 – 10:05	30 min	Utility & Advanced Technologies	**Jeff Mocha**, COO, Oakville Hydro
10:05 – 10:35	30 min	AI / ML Platform & Services for Industries	**Kushal Deshpande**, Technology Architect, AWS
10:35 – 11:05	30 min	AI: Governance and Guardrails - Trust	**Alastair Taylor**, Senior Strategic Advisor, Invitas
11:05 – 11:50	45 min	Themis: Human-Guided Intelligence	**Young Ngo**, President **Peter Jang**, Technology Director
11:50 – 12:00	10 min	Q&A, Closing Remarks	**All**

https://github.com/IIB-Lab/AI-ML-bootcamp-tutorials

UNIVERSITY OF TORONTO | Engineering

Unsupervised Machine Learning

Unsupervised Learning

- Dimensionality Reduction
  - PCA – linear; for features & visualization
  - t-SNE – nonlinear; for visualization
- Clustering
  - *k*-means
  - Gaussian mixture
- Many others…

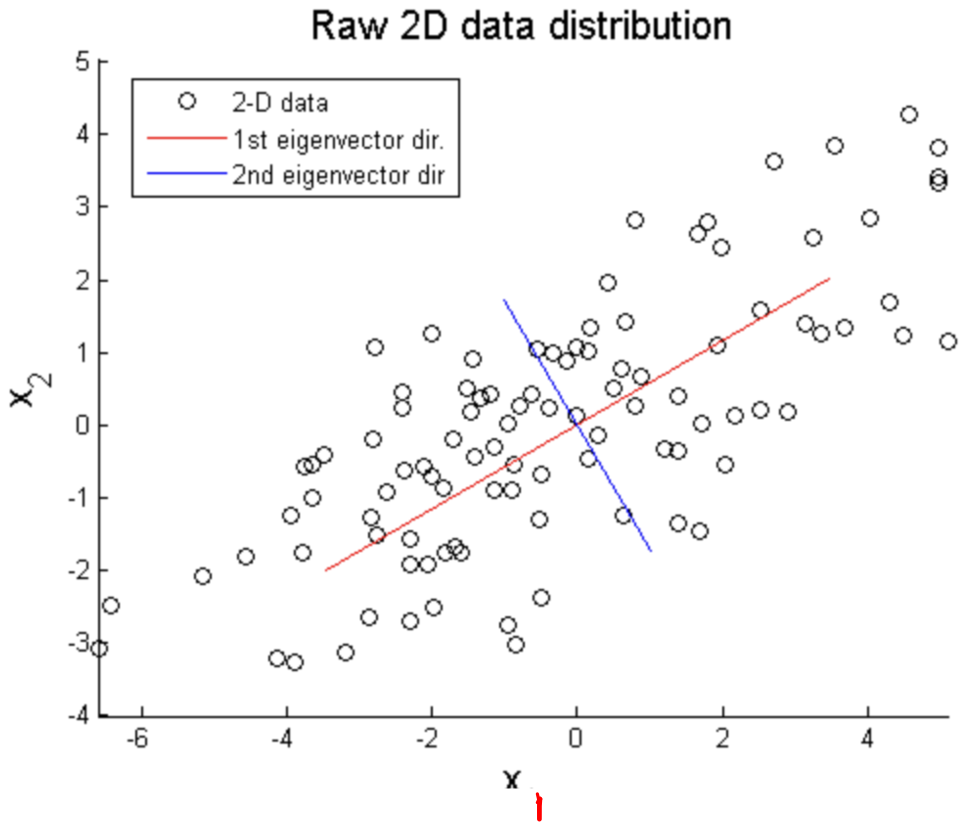
Dimensionality Reduction

- Complexity of model ~ dimensionality (# of features)
  - Dimensionality vs. sample size
  - Simpler models have lower variance
  - Simpler models are more explainable

- Reducing dimensionality
  - Feature selection
  - Feature extraction

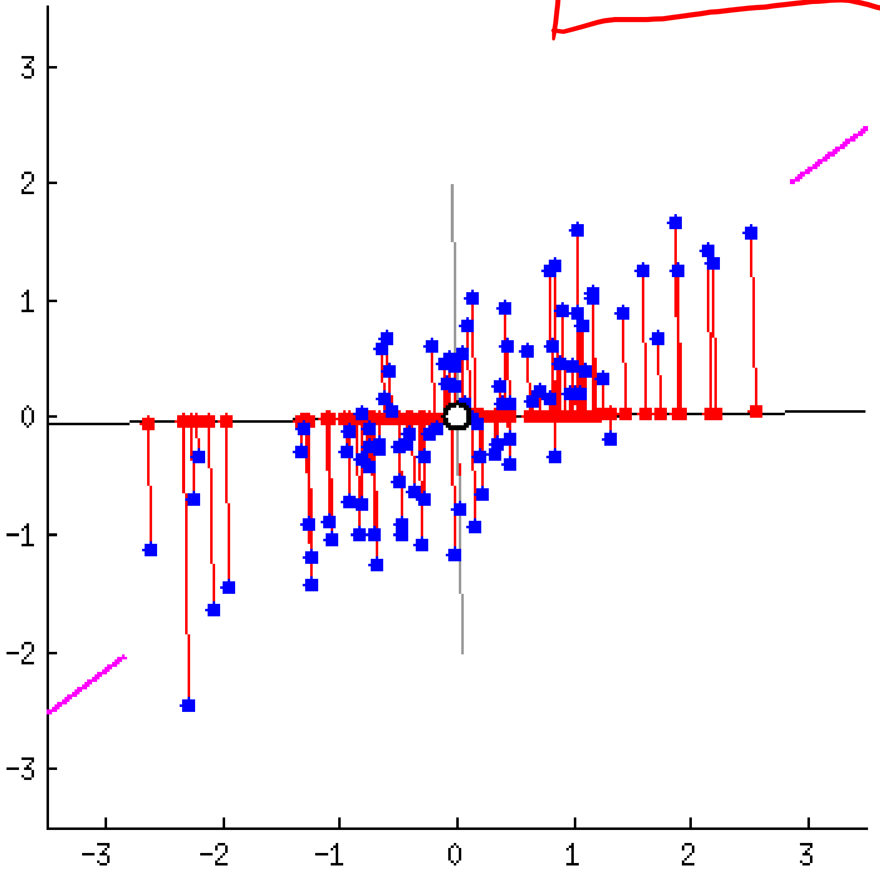
Feature Extraction

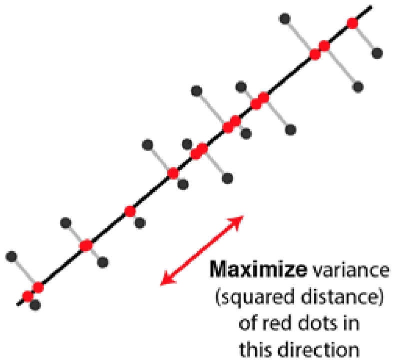
- PCA (Principal Component Analysis)
  - Dimensionality reduction technique
  - Also useful as a visualization tool
  - Significance of data should remain
    - Maximize the variance in the reduced space
    - Minimization of projection error



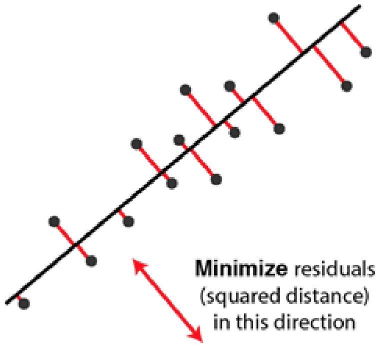
Feature Extraction

- PCA (Principal Component Analysis)
  - Dimensionality reduction technique
  - Also useful as a visualization tool
  - Significance of data should remain
    - Maximize the variance in the reduced space
    - Minimization of projection error

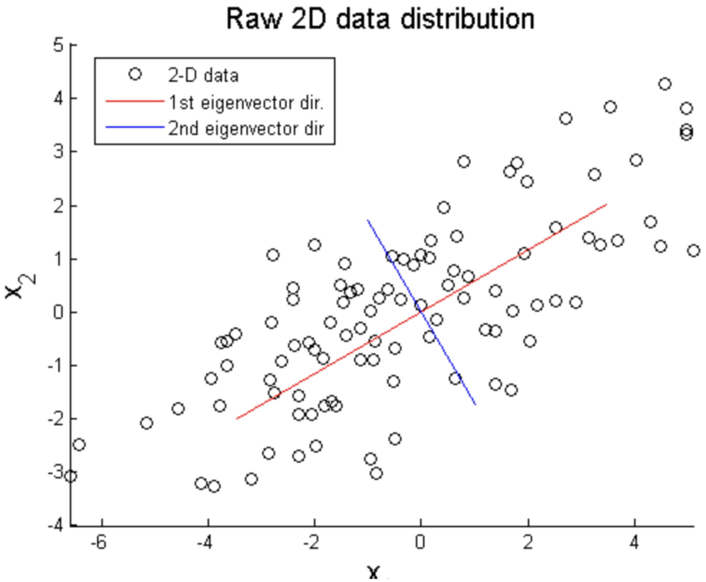




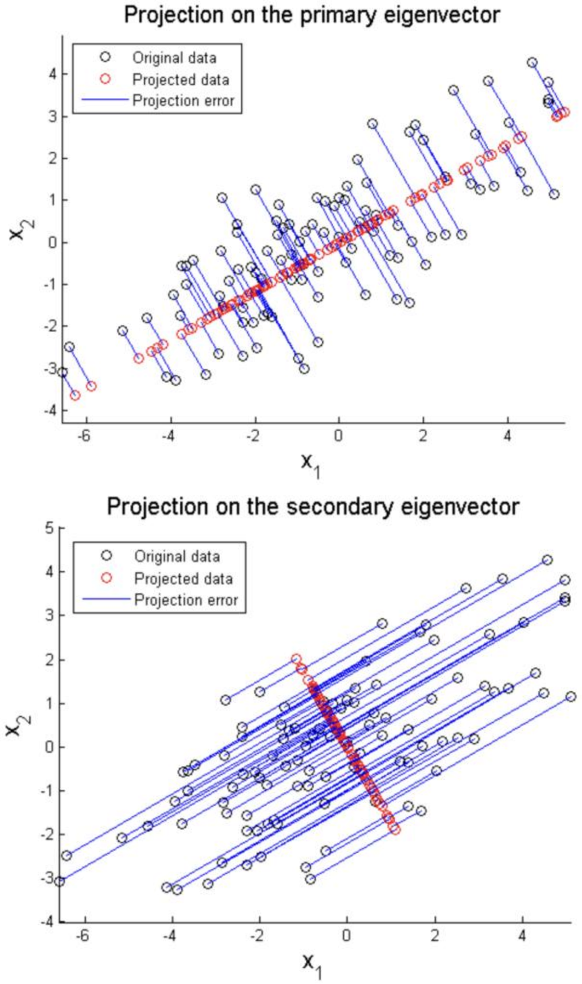
vs



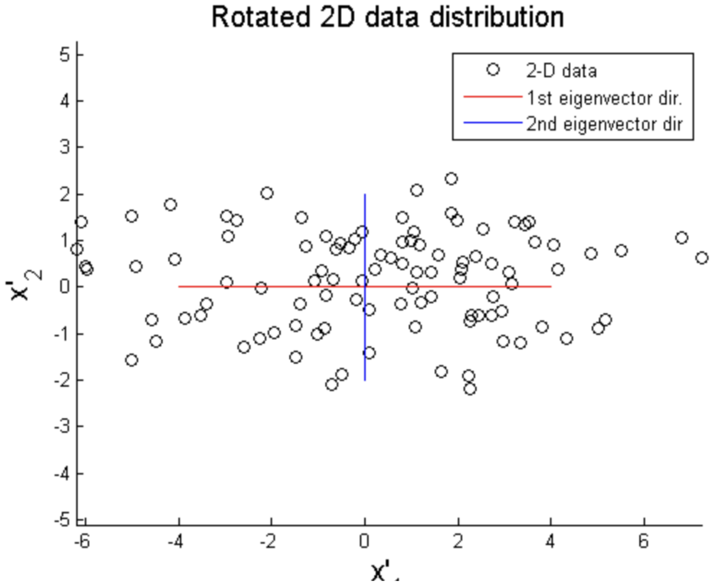
- Example



Raw 2D data distribution
2-D data
1st eigenvector dir.
2nd eigenvector dir



Projection on the primary eigenvector
Original data
Projected data
Projection error
Projection on the secondary eigenvector
Original data
Projected data
Projection error



Rotated 2D data distribution
2-D data
1st eigenvector dir.
2nd eigenvector dir

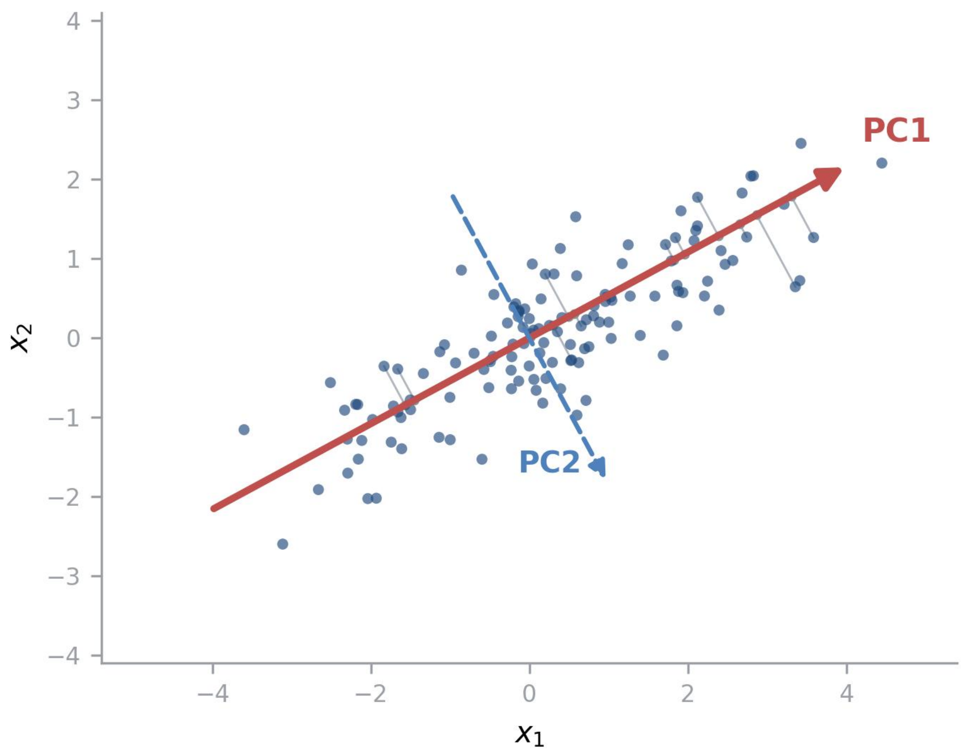
PCA as an Optimization Problem

- Project each point x onto a direction w: $z = w^{\mathrm{T}} x$
- Keep as much information as possible → maximize the variance of z
- The objective:

$$\max_w \; w^{\mathrm{T}} \Sigma w \quad \text{subject to} \quad \|w\| = 1$$

(without the constraint, variance would grow just by scaling w)

- Solved in closed form → next slide



PC1: the direction along which the data varies most

PCA: From Optimization to Eigenvectors

$Av = \lambda v$

- Solving with Lagrange multipliers gives a clean answer

$$\Sigma w = \lambda w$$

the best directions are eigenvectors of the covariance matrix Σ

$w^T \Sigma w = \lambda w^T w = \lambda \quad \Rightarrow \quad$ each eigenvalue = variance captured by its PC

- Sort eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots$: PC1 = top eigenvector, PC2 = next (orthogonal to PC1), …
- $\lambda_i / (\lambda_1 + \lambda_2 + \cdots)$ = fraction of variance explained → how many components to keep

Steps of PCA

1. Standardization/normalization of Data
2. Computation of Covariance Matrix Σ
3. Computation of eigenvector and eigenvalues
4. Computation of Principal Components
5. Projection of the data using the principal components vectors

Steps of PCA

- Data and Standardization

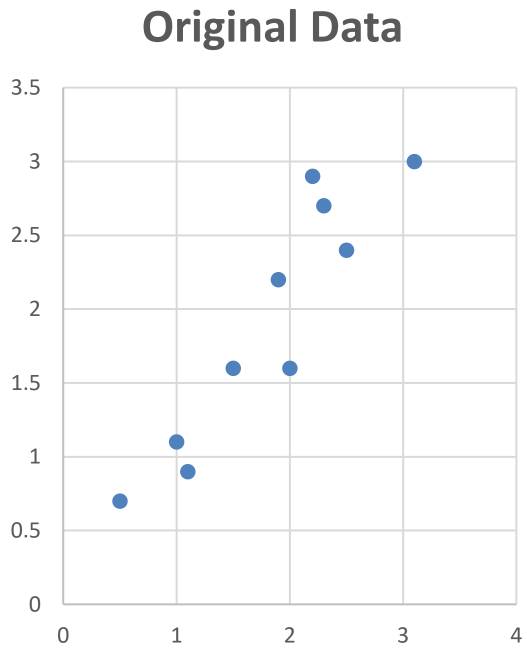
Data =

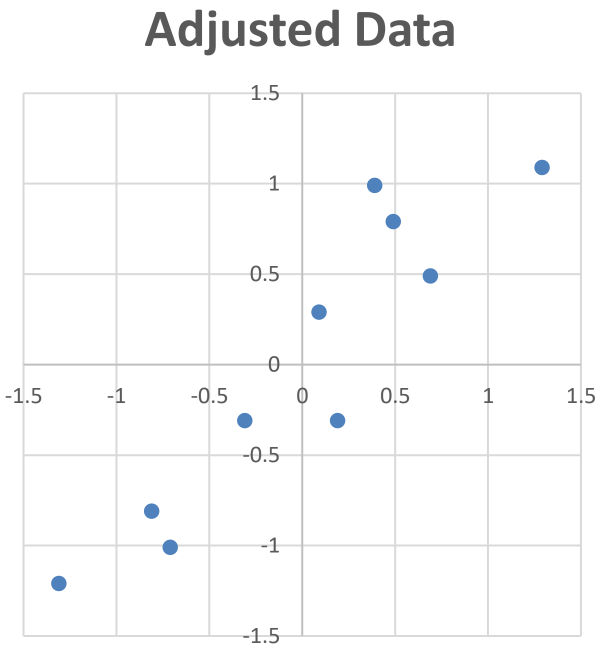
x	y
2.5	2.4
0.5	0.7
2.2	2.9
1.9	2.2
3.1	3.0
2.3	2.7
2	1.6
1	1.1
1.5	1.6
1.1	0.9

$$(x', y') = (x - \bar{x}, y - \bar{y})$$

DataAdjust =

x	y
.69	.49
-1.31	-1.21
.39	.99
.09	.29
1.29	1.09
.49	.79
.19	-.31
-.81	-.81
-.31	-.31
-.71	-1.01





Steps of PCA

- Covariance matrix Σ

$$\Sigma = \begin{pmatrix} .616555556 & .615444444 \\ .615444444 & .716555556 \end{pmatrix}$$

x	y
.69	.49
-1.31	-1.21
.39	.99
.09	.29
1.29	1.09
.49	.79
.19	-.31
-.81	-.81
-.31	-.31
-.71	-1.01

	(X'-Avg(X))	Y'-Avg(Y)	(X'-Avg(X))(Y'-Avg(Y))
	0.69	0.49	0.3381
	-1.31	-1.21	1.5851
	0.39	0.99	0.3861
	0.09	0.29	0.0261
	1.29	1.09	1.4061
	0.49	0.79	0.3871
	0.19	-0.31	-0.0589
	-0.81	-0.81	0.6561
	-0.31	-0.31	0.0961
	-0.71	-1.01	0.7171
Sum			
Sum of squares	5.549	6.449	5.539
Sample Variance	**0.61655556**	**0.71655556**	**0.615444444**

Steps of PCA

- Eigenvalues

Given $\Sigma = \begin{pmatrix} .616555556 & .615444444 \\ .615444444 & .716555556 \end{pmatrix}$

We want to find λ such that $\det(\Sigma - \lambda I) = 0$

$$\begin{vmatrix} 0.617 - \lambda & 0.615 \\ 0.615 & 0.717 - \lambda \end{vmatrix} = 0$$

$$(0.617 - \lambda)(0.717 - \lambda) - 0.615^2 = 0$$

Steps of PCA

- Given eigenvalues, let's find eigenvectors $\boldsymbol{v}$

$$\Sigma v = \lambda v$$

$$(\Sigma - \lambda I)v = 0$$

For $\lambda = 1.28402771$

$$\begin{bmatrix} -0.677873399 \\ -0.735178656 \end{bmatrix}$$

For $\lambda = 0.0490834$

$$\begin{bmatrix} -0.735178656 \\ 0.677873399 \end{bmatrix}$$

Steps of PCA

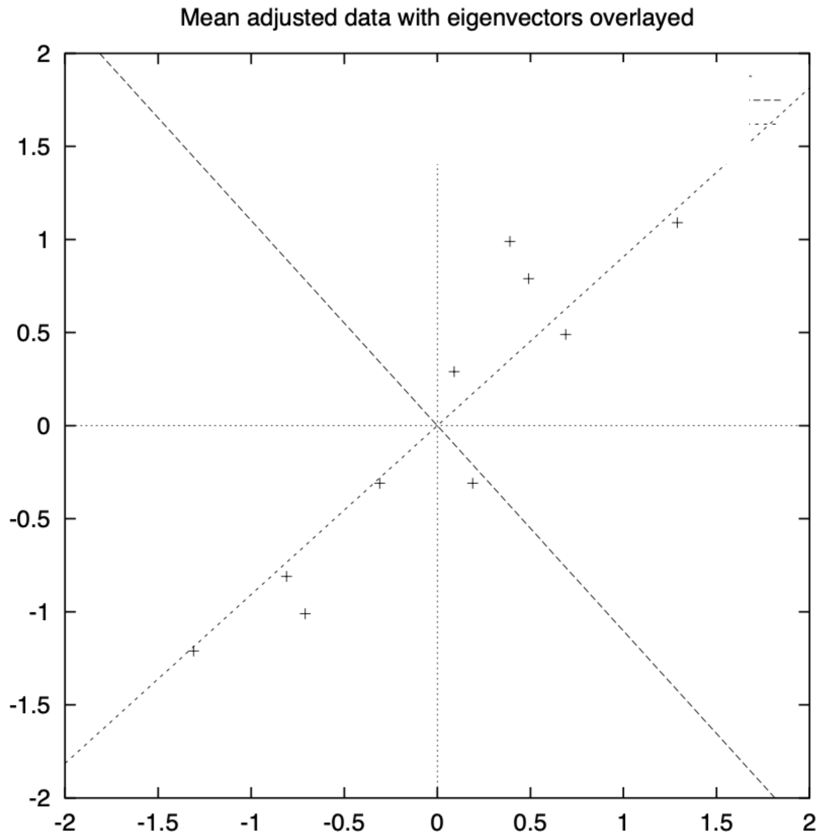
- Eigenvectors

$$\begin{bmatrix}-0.677873399\\-0.735178656\end{bmatrix} \qquad \begin{bmatrix}-0.735178656\\0.677873399\end{bmatrix}$$

- Feature Vector

$$\begin{bmatrix}-0.677873399\\-0.735178656\end{bmatrix}$$

$$\begin{bmatrix}-0.677873399, & -0.735178656\\-0.735178656, & 0.677873399\end{bmatrix}$$



Steps of PCA

- Projected Data

DataAdjust =

x	y
.69	.49
-1.31	-1.21
.39	.99
.09	.29
1.29	1.09
.49	.79
.19	-.31
-.81	-.81
-.31	-.31
-.71	-1.01

$$(\tilde{x}, \tilde{y}) = Feacture\ Vector \cdot (x, y)$$

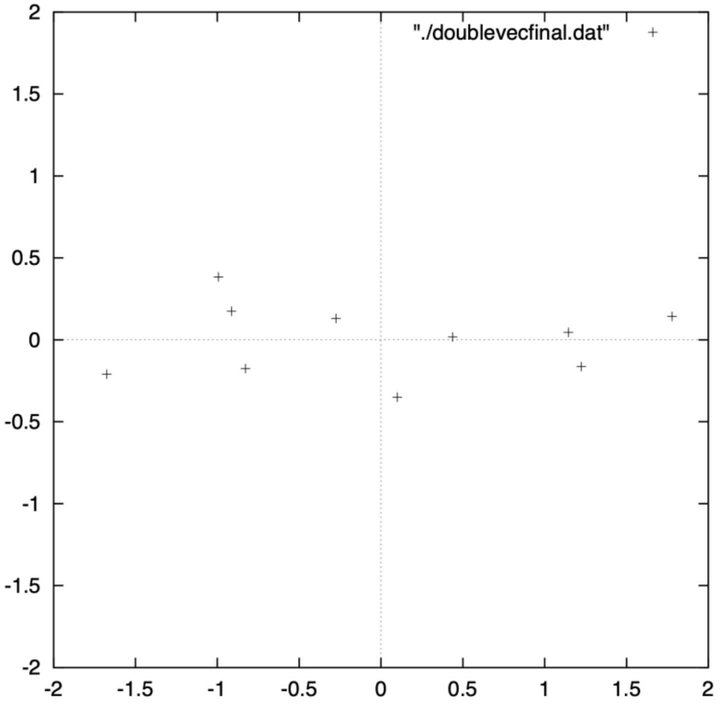
$\tilde{x}$
-.827970186
1.77758033
-.992197494
-.274210416
-1.67580142
-.912949103
.0991094375
1.14457216
.438046137
1.22382056

$$\begin{pmatrix} -.677873399 & -.735178656 \\ -.735178656 & .677873399 \end{pmatrix}$$

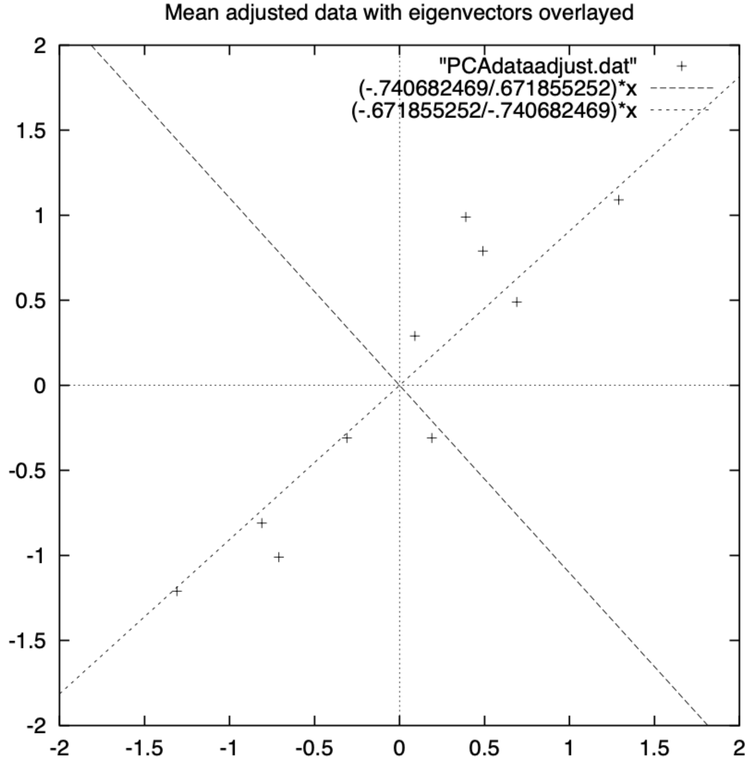
$\tilde{x}$	$\tilde{y}$
-.827970186	-.175115307
1.77758033	.142857227
-.992197494	.384374989
-.274210416	.130417207
-1.67580142	-.209498461
-.912949103	.175282444
.0991094375	-.349824698
1.14457216	.0464172582
.438046137	.0177646297
1.22382056	-.162675287

Steps of PCA

x
-.827970186
1.77758033
-.992197494
-.274210416
-1.67580142
-.912949103
.0991094375
1.14457216
.438046137
1.22382056

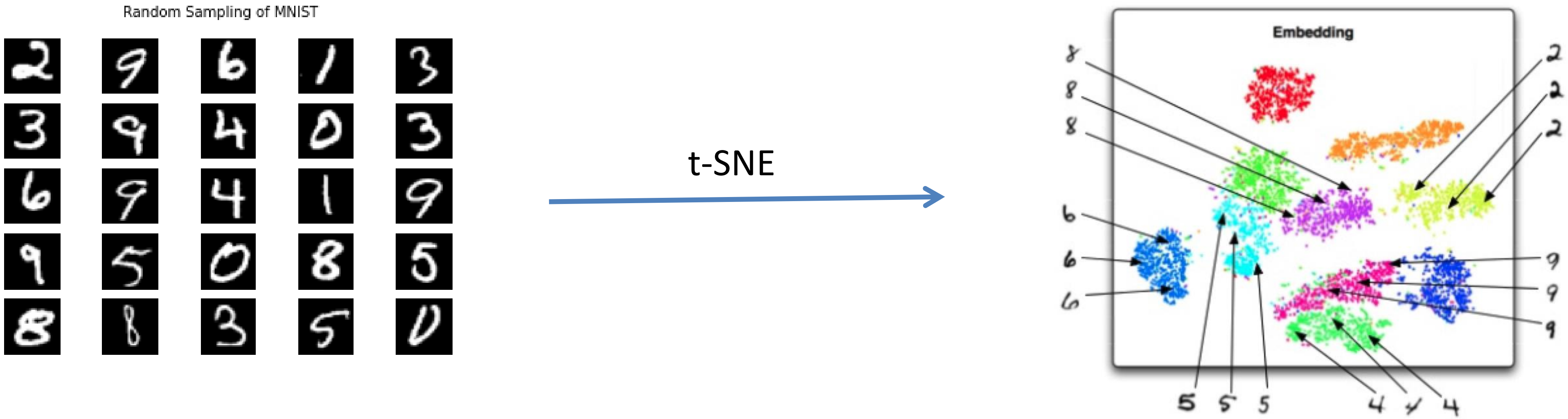


x	y
-.827970186	-.175115307
1.77758033	.142857227
-.992197494	.384374989
-.274210416	.130417207
-1.67580142	-.209498461
-.912949103	.175282444
.0991094375	-.349824698
1.14457216	.0464172582
.438046137	.0177646297
1.22382056	-.162675287



t-SNE

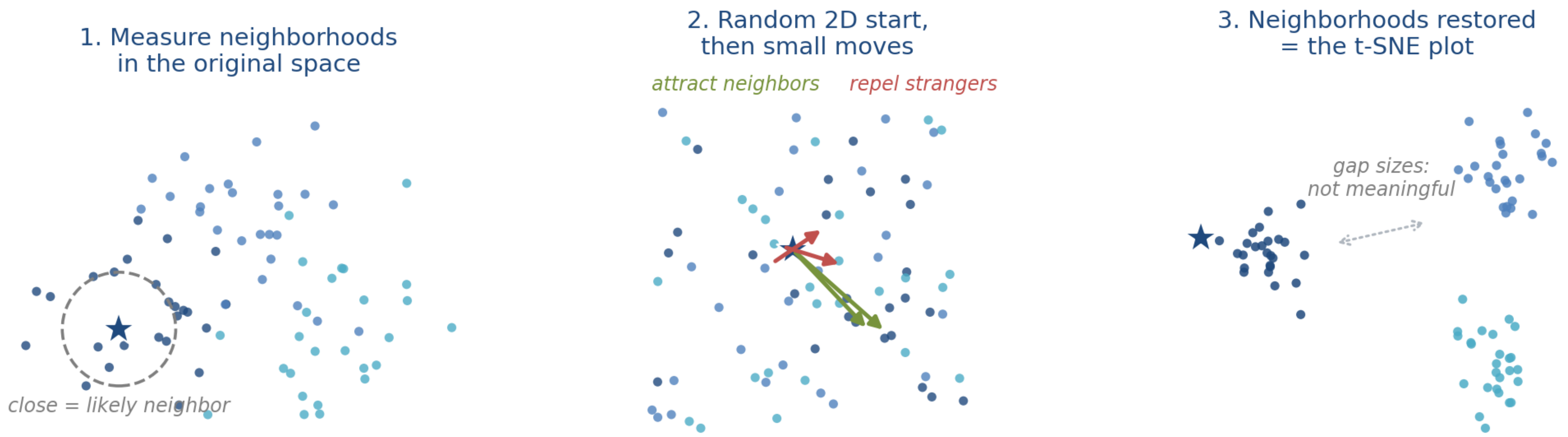
- t-Distributed Stochastic Neighbor Embedding



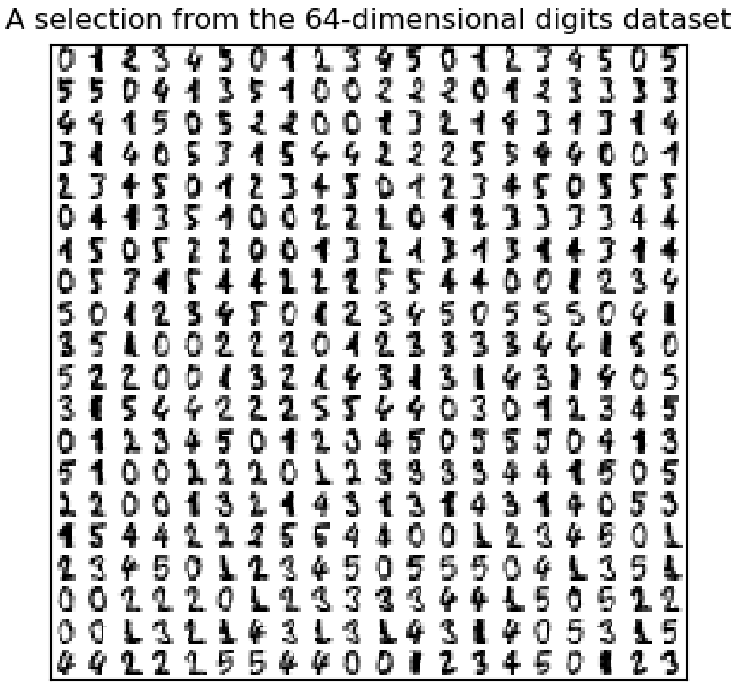
  – Stochastic tool to retain local relationship through dimensionality reduction
  – Distance between clusters is not controlled
  – Computationally expensive

t-SNE: How It Works

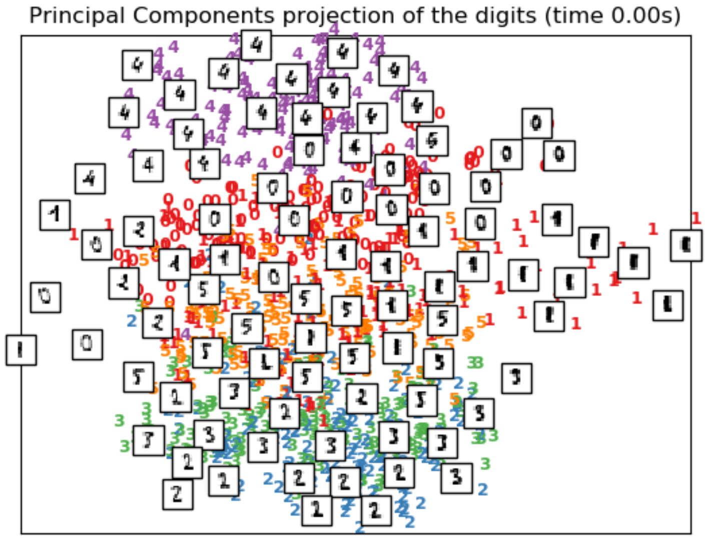
- **Step 1:** ask each point "who are my neighbors?" — close points get high neighbor probability
- **Step 2:** place all points in 2D at random, and ask the same question there
- **Step 3:** nudge points until the 2D answers match the originals — neighbors attract, strangers repel
- **The "t":** heavy-tailed distribution in 2D gives clusters extra room — hence the well-separated blobs



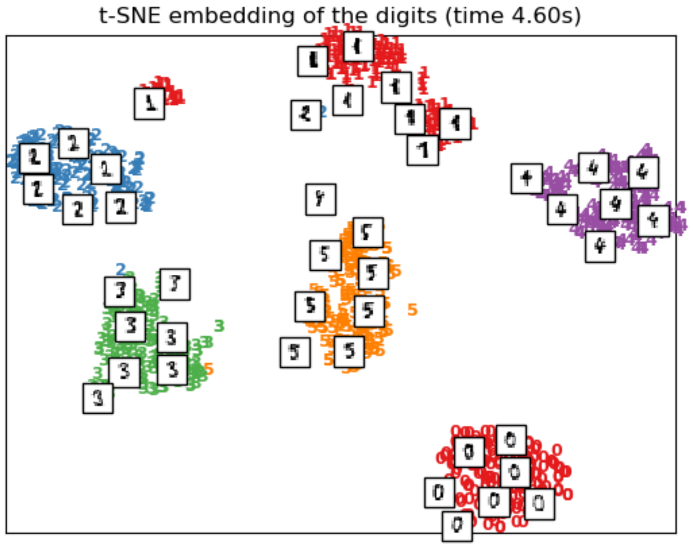
t-SNE vs. PCA



a. PCA



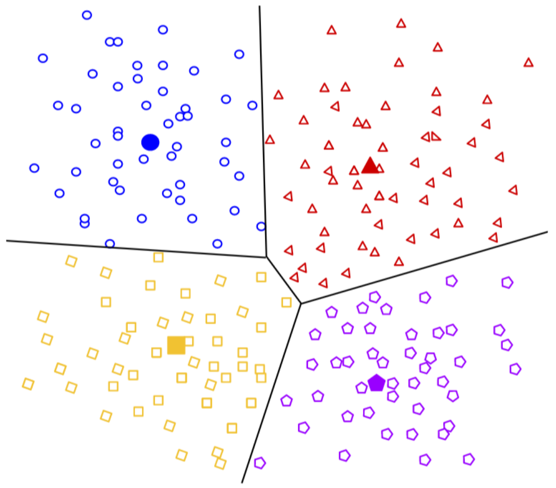
b. t-SNE



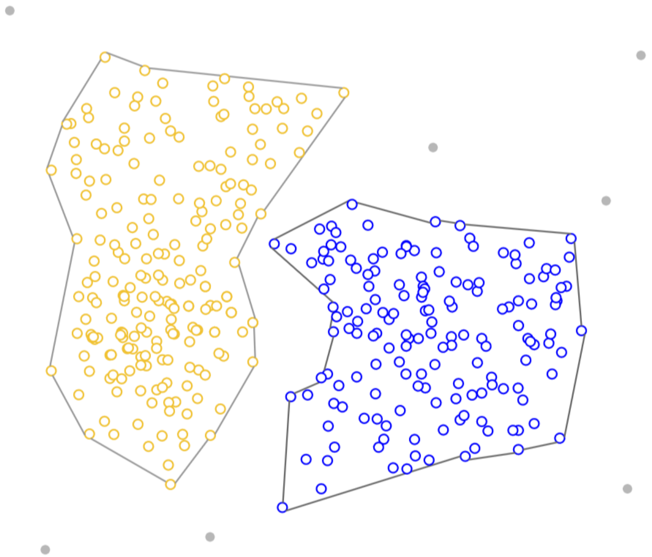
Clustering

- A process of grouping a set of data objects into groups or clusters
  - So that objects within a cluster have high similarity,
  - but are dissimilar to objects in other clusters.
- An unsupervised ML
- Applications
  - Anomaly detection
  - Marketing (segmentation)
  - Recommender system
  - Social network analysis

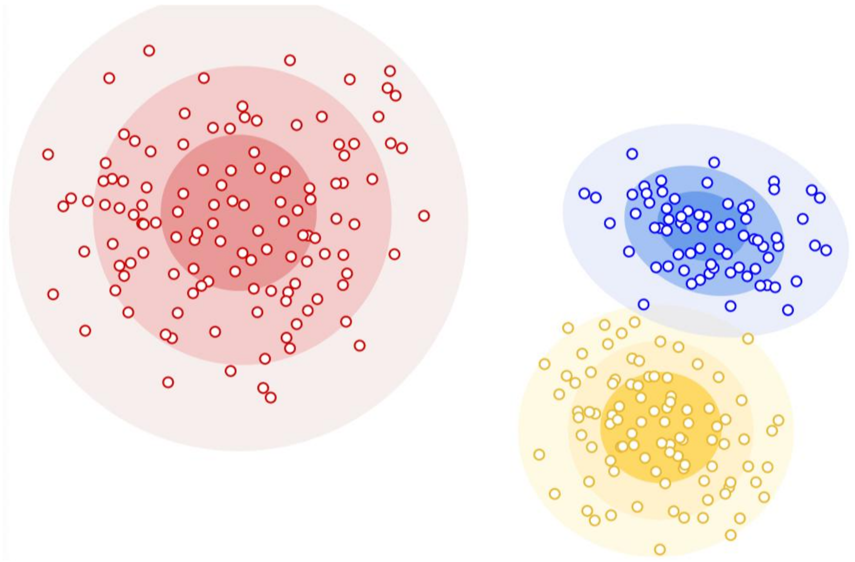
Clustering



Centroid-based



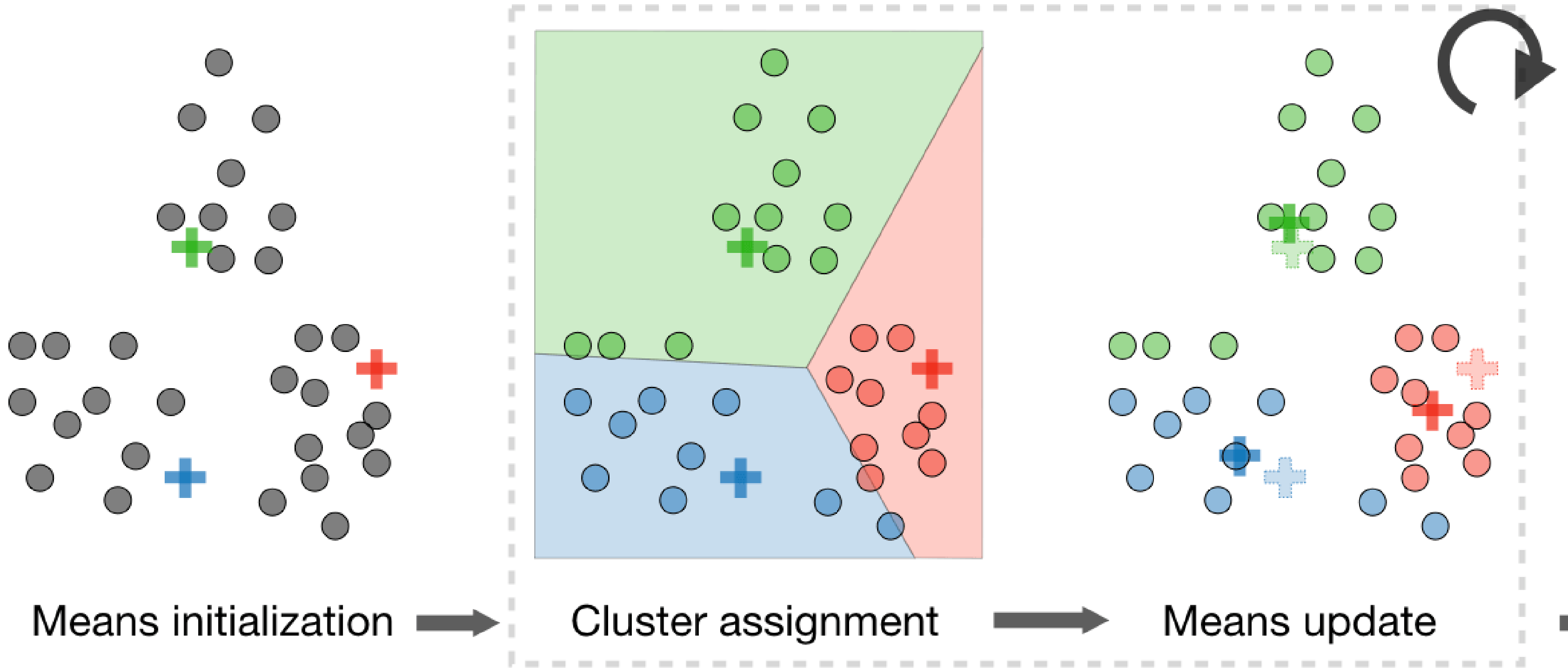
Density-based



Distribution-based

Hard Clustering

- k-Means Clustering

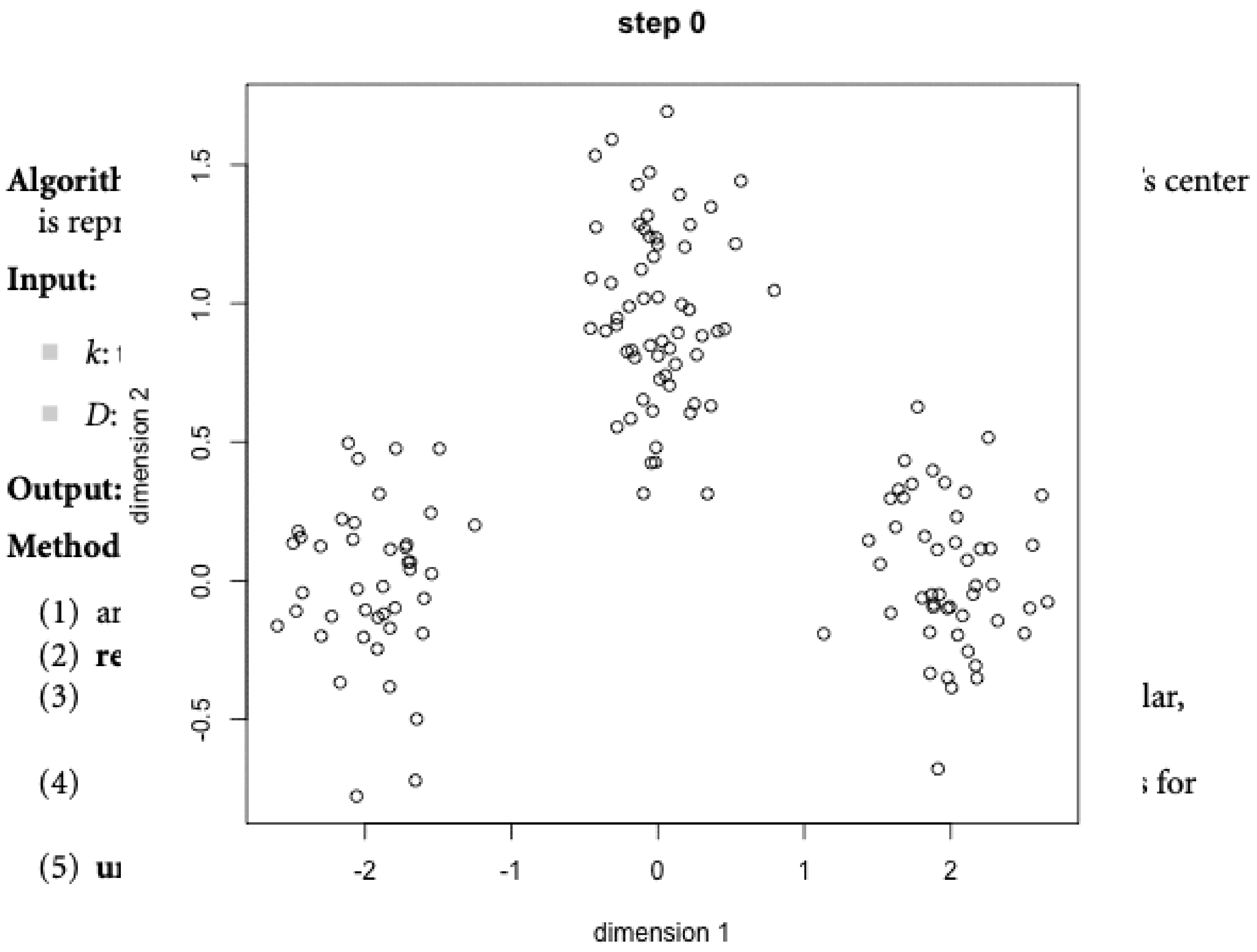


k-Means Clustering

- Error

$$E = \sum_{i=1}^{k} \sum_{p \in C_i} (p - c_i)^2$$

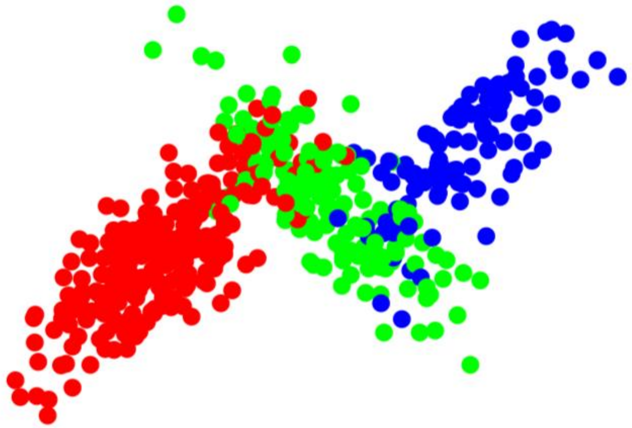
- Drawbacks
  - No guarantee for convergence to the global optimum
  - Final results depend on the initial random selection of cluster centres
  - Sensitive to outliers (k-Medoids)

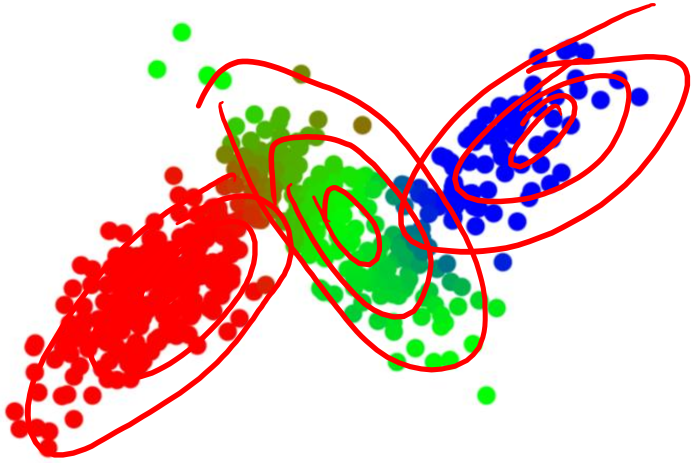


- Scale your features before k-means

Gaussian Mixtures

Steps	k-means	Gaussian Mixture
Parameter computation	Compute new **centroids** given hard allocation	Compute **mean & variance** of component Gaussian distributions given soft allocation
Cluster allocation	Assign samples to the nearest centroid (hard allocation)	Assign samples to all clusters with fractional membership (soft allocation)





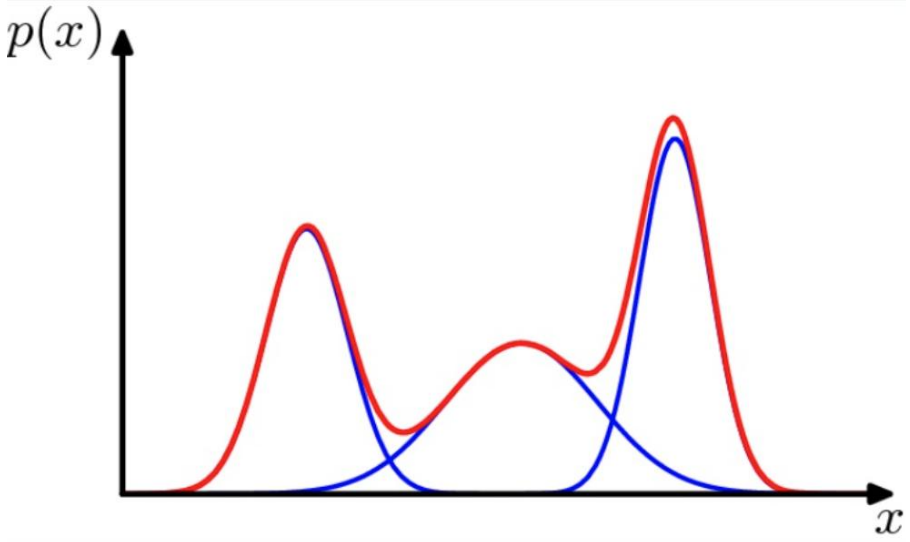
Gaussian Mixtures

- Clustering ≡ Mixture Distribution Fitting
- Data distribution can be

$$p(\mathbf{x}) = \sum_{k=1}^{K} \pi_k \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k).$$

cluster membership (responsibilities)

data distribution in cluster k



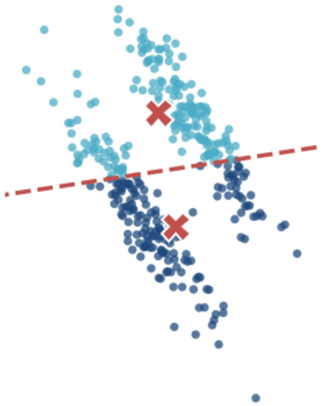
- Log likelihood can be maximized
  - Usually, it is intractable
  - Alternatively find membership and distribution in each cluster (Expectation Maximization Alg)

k-Means vs. Gaussian Mixtures

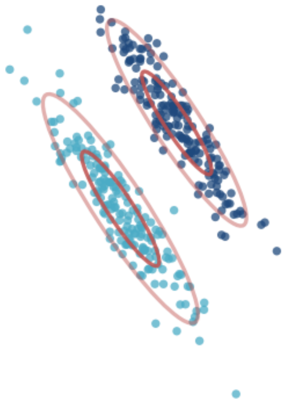
	k-Means	Gaussian Mixture
Assignment	Hard — each point joins exactly one cluster	Soft — a probability for every cluster
Cluster shape	Spherical, similar sizes (distance only)	Elliptical — any size and orientation
Algorithm	Assign points → update centroids (repeat)	E-step → M-step (repeat) — the same rhythm
You get	Labels — fast; the default first try	Probabilities — slower, more flexible

k-means ≈ a Gaussian mixture with identical, spherical clusters and hard assignments — both need k; k-means also needs scaled features

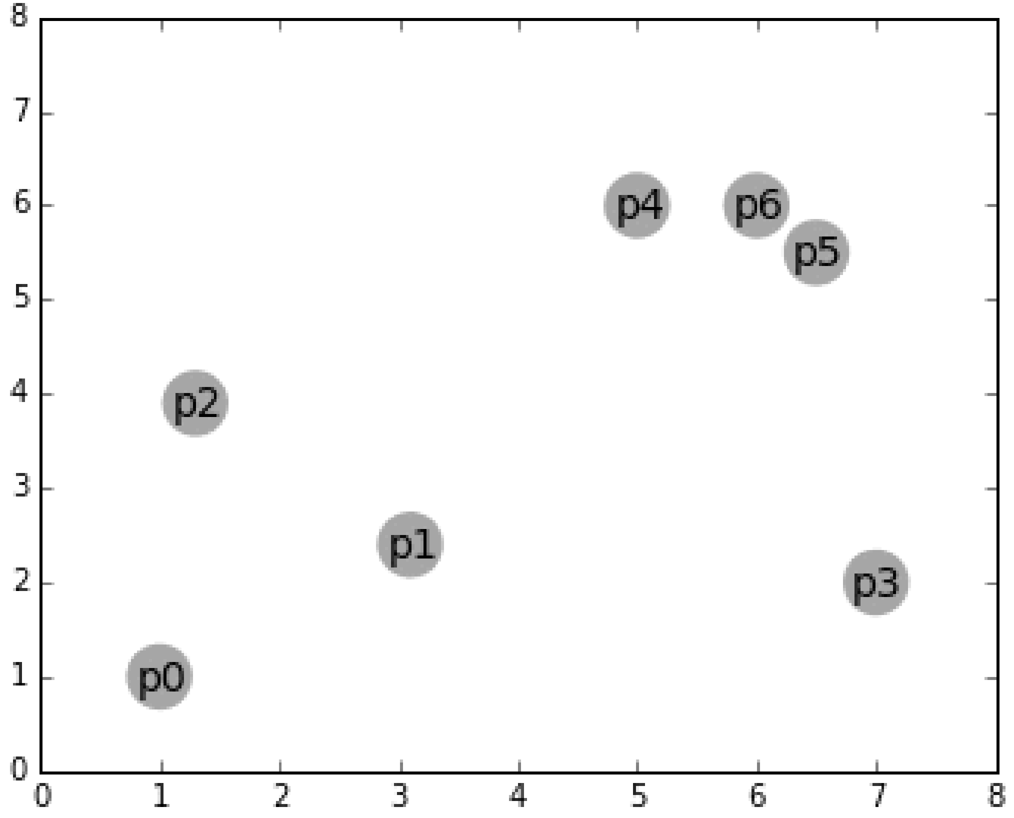
k-Means: spherical assumption → wrong split

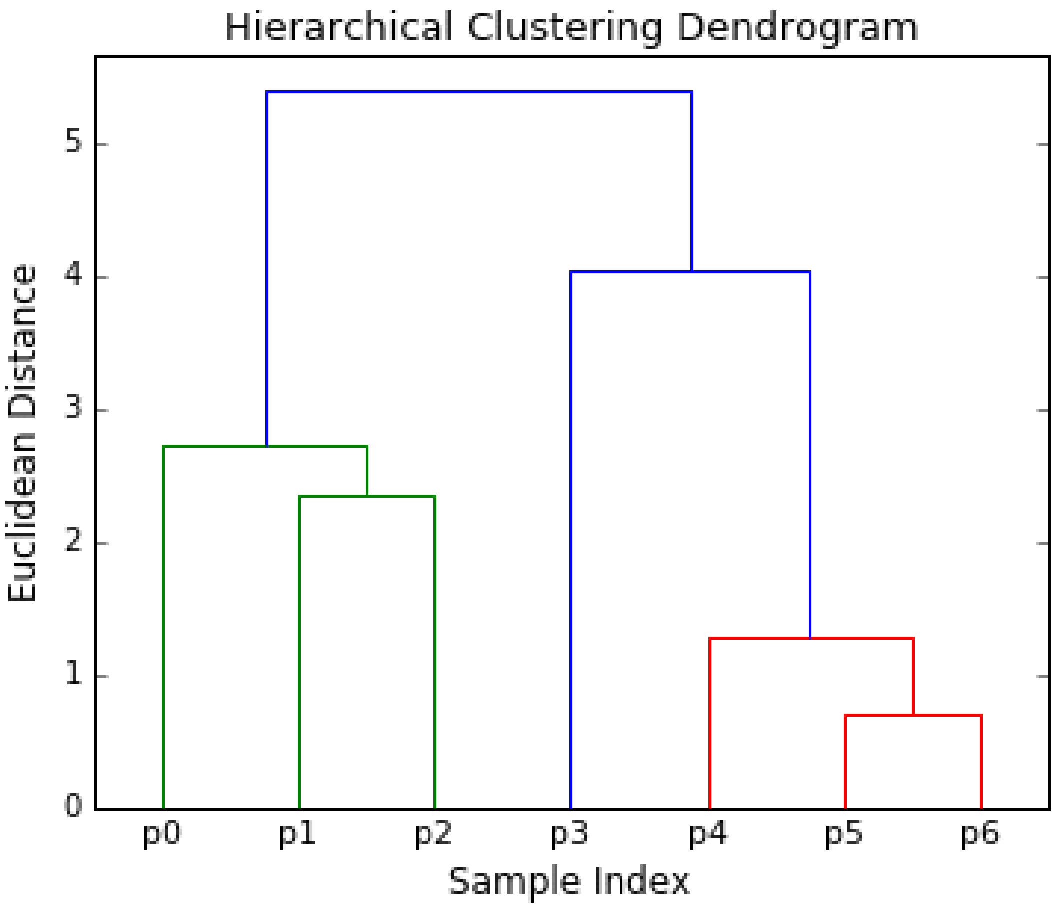


Gaussian Mixture: ellipses follow the shape



Other Clustering Algorithms: Hierarchical





Clustering

- Determining the # of Clusters
  - To balance between compressibility & accuracy
  - Depends on distribution's shape, scale of the data set, required resolution
  - Simple Rule

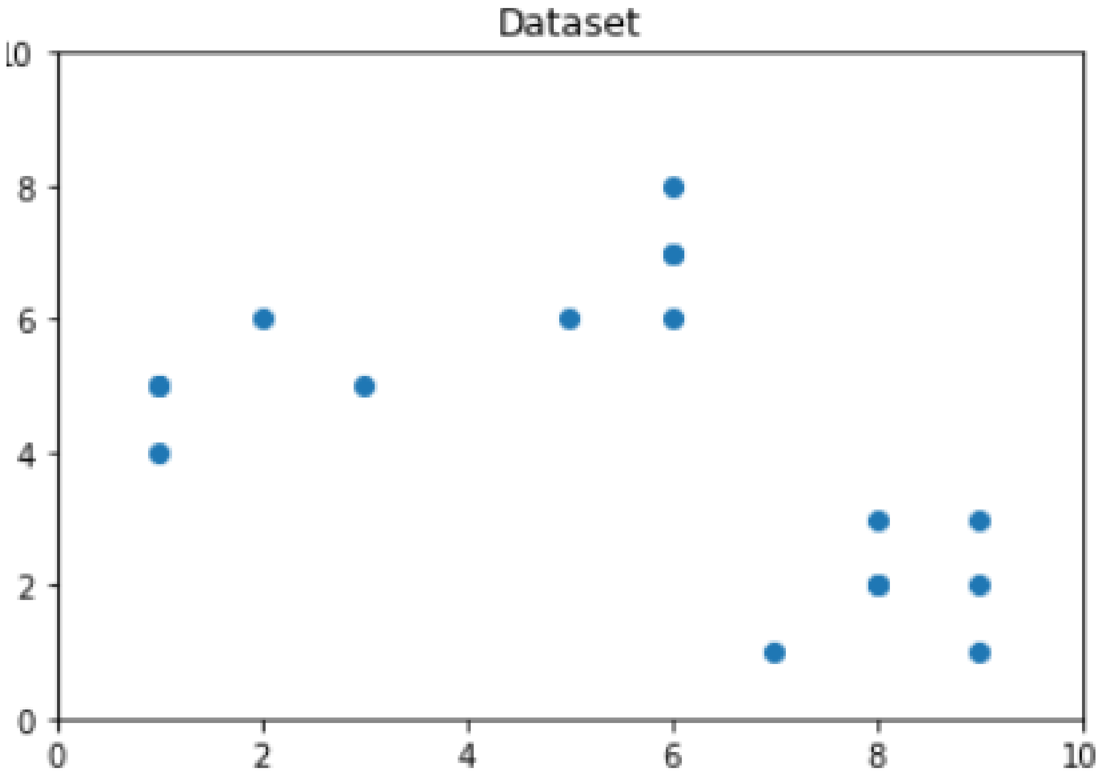
$$k = \sqrt{\frac{n}{2}}$$

  - In which case each cluster has $\sqrt{2n}$ points

  - The Elbow Method
    - More clusters means smaller within-cluster variance
    - Marginal variance reduction decreases
    - Hence, we can heuristically look for a turning point

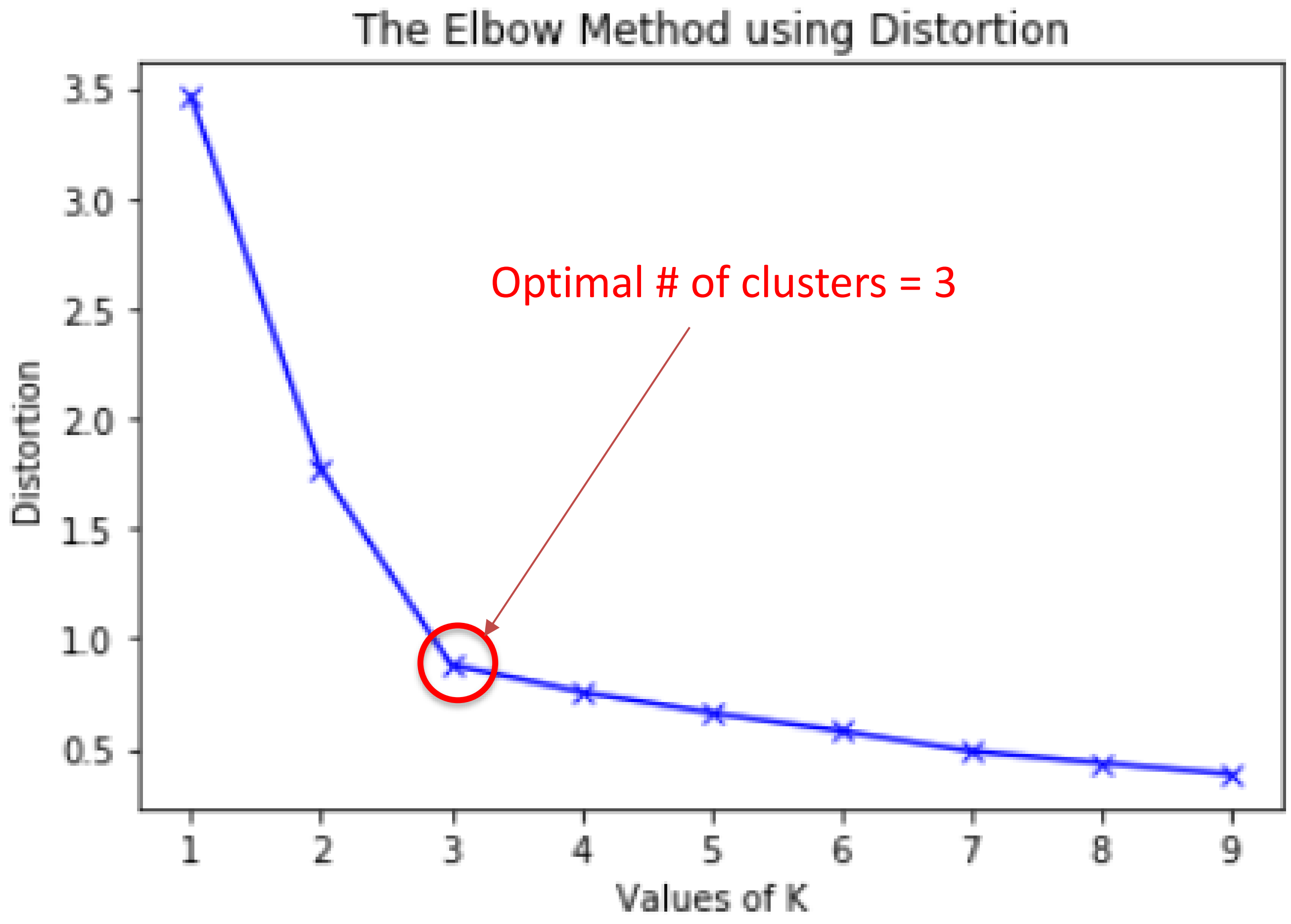
```
1: Form k clusters
2: Compute the sum of within-
   cluster error (k)
3: Plot the curve of error w.r.t. k
4: Find the first turning point in
   the curve
```

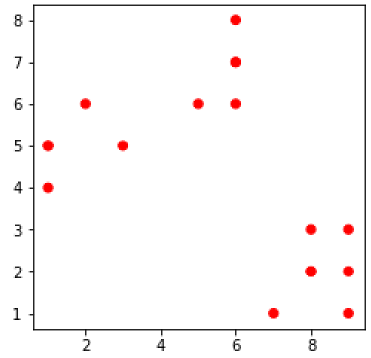
The Elbow Method



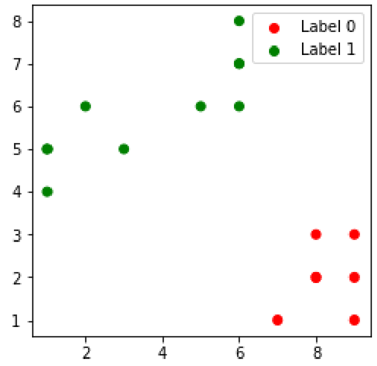
k	Sum of inner cluster variance
1	3.4577032384495707
2	1.7687413573405673
3	0.8819889697423957
4	0.7587138847606585
5	0.6635212812400347
6	0.5808803063754726
7	0.5093717077076824
8	0.41652236641410356
9	0.3333333333333333

The Elbow Method

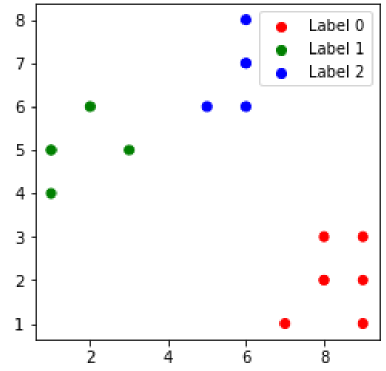




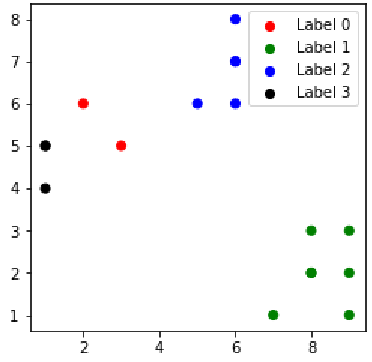
# of clusters=1



# of clusters=2



# of clusters=3



# of clusters=4

Case Study

Data

- 480 generating units
  - Hydro power plant in Niagara Falls, Canada
  - Over 5 years (2012~2017)
- The units failed for various causes from 114 components.
- 0.6 million entries of maintenance records, failures, etc.

472-megawatt steam turbine generator (photo credit: businesswire.com)

Data Collection

1. Removing redundancy
2. Removing units with inadequate records
3. Removing or recovering incomplete/inconsistent observations

	ForceOut	MainOut	MaxCapability	NumofCommon	PlanOut	WorkingHour	G199999	G142100	G141100	G142115	...
HGU0001	11	13	77.0	0.0	12	33429.6	0.0	6.0	2.0	0.0	...
HGU0002	13	14	77.0	0.0	10	36574.6	0.0	9.0	1.0	0.0	...
HGU0003	5	14	77.0	0.0	6	35721.3	0.0	3.0	1.0	0.0	...
HGU0004	11	10	77.0	0.0	7	36983.2	0.0	4.0	0.0	1.0	...
HGU0005	11	13	77.0	0.0	9	39080	0.0	6.0	1.0	0.0	...
HGU0006	6	13	77.0	0.0	8	35225.6	0.0	6.0	2.0	0.0	...
HGU0007	5	4	150.0	0.0	7	40213.8	1.0	7.0	16.0	1.0	...

Clustering Results

	Cluster 1	Cluster 2	Cluster 0
Average number of Forced outages	25.985	13.603	17.460
Average number of Maintenance outages	30.758	15.026	23.400
Average number of Planned outages	15.833	10.250	26.100
Average number of Common modes	0.015	1.263	0.280
Average maximum capability	46.533	58.185	306.586
Average working hours	37738.700	38917.044	35084.975

Clustering Results

Cluster 0

- Largest average maximum capacity/unit
- Medium reliability
- Highest planned outages number is scheduled on these units
- Cluster 0 seems mostly important to the company.

Cluster 1

- Smallest average maximum capability/unit
- The least reliable units
- Relatively large number of planned outages
- Failures seem isolated rather than caused by other units

Cluster 2

- The most reliable unit
- Outages are likely caused by other units (leading to correlation analysis)
- Least maintained

UNIVERSITY OF TORONTO | Engineering

Thank you.

Appendix

Principal Component Analysis

- Projection of x on the direction of w

$$z = w^T x$$

- Maximizing the variability subject to size of vector = 1

$$\max_{w_1} \; Var(z_1) = w_1^T \Sigma w_1$$

$$\text{s.t.} \quad w_1^T w_1 = 1$$

- By Lagrangian

$$\max_{w_1} \; w_1^T \Sigma w_1 - \alpha(w_1^T w_1 - 1)$$

- By first order condition

$$2\Sigma w_1 - 2\alpha w_1 = 0 \qquad \text{or} \qquad \Sigma w_1 = \alpha w_1$$

Principal Component Analysis

- We have

$$\Sigma w_1 = \alpha w_1$$

  - w_1 is an eigenvector of Σ and α is the corresponding eigenvalue

- Since we try to maximize $w_1^T \Sigma w_1 = \alpha w_1^T w_1 = \alpha$, we choose the eigenvector with the **largest eigenvalue**

- **The second PCA** w_2 should also maximize the variance, be of unit length and be orthogonal to w_1

$$\max_{w_2} \; w_2^T \Sigma w_2 - \alpha(w_2^T w_2 - 1) - \beta(w_2^T w_1 - 0)$$

- By first order condition

$$2\Sigma w_2 - 2\alpha w_2 - \beta w_1 = 0$$

- Multiplying w_1

$$2w_1^t \Sigma w_2 - 2w_1^t \alpha w_2 - \beta w_1^t w_1 = 0$$

Principal Component Analysis

- From the previous slide,

$$2w_1^T\Sigma w_2 - 2w_1^T\alpha w_2 - \beta w_1^T w_1 = 0 \tag{1}$$

- Since

$$w_1^T w_2 = 0 \qquad\qquad \Sigma w_1 = \lambda_1 w_1$$

- $w_1^T\Sigma w_2$ is a scalar and hence its transpose has the same the value (i.e., $w_1^T\Sigma w_2 = w_2^T\ \Sigma w_1$). Therefore,

$$w_1^T\Sigma w_2 = w_2^T\ \Sigma w_1 = \lambda_1 w_2^T w_1 = 0$$

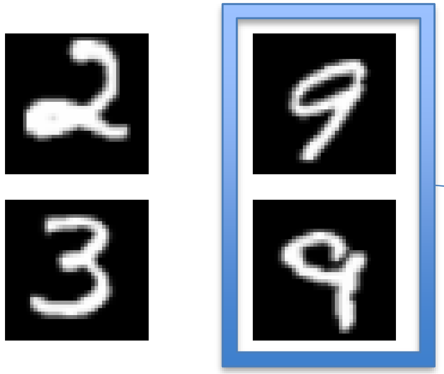
- (1) becomes $0 - 0 - \beta \cdot 1 = 0$ implying $\beta = 0$. As a result,

$$2\Sigma w_2 - 2\alpha w_2 - \beta w_1 = 0 \quad \text{becomes} \quad \Sigma w_2 = \alpha w_2$$

- Similar to the first PCA, the second PCA is the eigenvector of Σ with the **second largest eigenvalue**

t-SNE

- PCA: Push points away from each other
- t-SNE: Keep closer points closer



- Nearness measured by probability that x_i belongs to the neighborhood of x_j than others

$$p_{i|j} := \frac{\exp(-|x_i - x_j|^2 / 2\sigma_j^2)}{\sum_{k \neq j} \exp(-|x_k - x_j|^2 / 2\sigma_j^2)}$$

- Keep the probability high in the reduced space

$$q_{ij} = \frac{\left(1 + \|y_i - y_j\|^2\right)^{-1}}{\sum_{k \neq l} \left(1 + \|y_k - y_l\|^2\right)^{-1}}$$

UNIVERSITY OF TORONTO | Engineering

t-SNE

- Steps
  1. Construct a probability distribution on pairs in higher dimensions

$$p_{ij} = \frac{p_{ji} + p_{ij}}{2N}$$

  2. Construct a probability distribution on pairs in the reduced dimensions

$$q_{ij} = \frac{\left(1 + \|y_i - y_j\|^2\right)^{-1}}{\sum_{k \neq l} \left(1 + \|y_k - y_l\|^2\right)^{-1}}$$

  3. Use SGD to minimize the discrepancy between the two distributions

$$\mathrm{KL}\left(P \parallel Q\right) = \sum_{i \neq j} p_{ij} \log \frac{p_{ij}}{q_{ij}}$$

EM for Gaussian Mixtures

1. Initialize the means μ_k, covariances Σ_k and mixing coefficient π_k
2. (**Expectation**) Compute the responsibilities

$$\gamma(z_{nk}) = \frac{\pi_k \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^{K} \pi_j \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}$$

3. (**Maximization**) Update the parameters given the current responsibilities

$$\boldsymbol{\mu}_k^{\text{new}} = \frac{1}{N_k} \sum_{n=1}^{N} \gamma(z_{nk}) \mathbf{x}_n \qquad \pi_k^{\text{new}} = \frac{N_k}{N}$$

$$\boldsymbol{\Sigma}_k^{\text{new}} = \frac{1}{N_k} \sum_{n=1}^{N} \gamma(z_{nk}) \left(\mathbf{x}_n - \boldsymbol{\mu}_k^{\text{new}}\right) \left(\mathbf{x}_n - \boldsymbol{\mu}_k^{\text{new}}\right)^{\mathrm{T}} \qquad N_k = \sum_{n=1}^{N} \gamma(z_{nk})$$

4. If converged, stop; otherwise, go to Step 2

Partial Derivations

- Responsibilities as a probability distribution
  - Let $z \in \{0,1\}^K$ such that $z_k \in \{0,1\}$ and $\sum_k z_k = 1$
  - Let $p(z_k = 1) = \pi_k$ $(0 \leq \pi_k \leq 1,\ \ \sum_{k=1}^{K} \pi_k = 1)$
  - We can write this distribution as

$$p(\boldsymbol{z}) = \prod_{k=1}^{K} \pi_k^{z_k}$$

- Conditional distribution of $\boldsymbol{x}$ given a particular value of $\boldsymbol{z}$

$$p(x|z_k = 1) = \mathcal{N}(x|\mu_k, \Sigma_k)$$

Hence,

$$p(\boldsymbol{x}|\boldsymbol{z}) = \prod_{k=1}^{K} \mathcal{N}(\boldsymbol{x}|\boldsymbol{\mu_k}, \boldsymbol{\Sigma_k})^{z_k}$$

- Posterior distribution of responsibilities z_k

$$\gamma(z_k) \equiv p(z_k = 1|x) = \frac{p(z_k = 1)p(x|z_k = 1)}{\sum_{j=1}^{K} p(z_j = 1)p(x|z_j = 1)}$$

$$= \frac{\pi_k \mathcal{N}(x|\mu_k, \Sigma_k)}{\sum_{j=1}^{K} \pi_j \mathcal{N}(x|\mu_j, \Sigma_j)}$$